

Introduction to Local Area Networks

A local area network, commonly called a LAN, connects computers, printers, servers, and other devices inside a limited geographic area such as an office, a laboratory, a classroom building, or a small campus. Unlike a wide area network, a LAN is owned and administered by a single organization. That organization controls cabling, switches, addressing policy, and access rules.

Typical LAN services include file sharing, printer sharing, directory authentication, software deployment, and access to internal web applications. Workstations act as clients, while dedicated machines provide file, print, mail, and database services. A peer-to-peer LAN is possible in a very small office, but most institutional networks follow a client-server model because it is easier to back up data and enforce security.

The physical reach of a LAN is short, so latency is low and bandwidth is high compared with the public Internet. Copper twisted-pair cable, fiber optic backbone links, and wireless access points are the usual media. A well designed LAN separates user access from server access and isolates laboratory equipment from administrative offices.

Network administrators plan capacity, monitor utilization, and keep an inventory of every switch port. They also maintain documentation: floor plans, patch-panel labels, IP address spreadsheets, and change logs. Without that documentation, troubleshooting a local area network becomes slow and error-prone.

In this collection the LAN is treated as the primary information space: documents describe topology, Ethernet, addressing, virtual LANs, wireless extensions, and security controls that protect hosts attached to the same local network.